

SYSTEM MONITOR TOOL

OBJECTIVE:- Create a system monitor tool in C++ that displays real time information about system processes, memory usage ,and CPU load ,similar to ‘TOP’ command.

DAY1: DESIGN UI LAYOUT AND GATHER SYSTEM DATA USING SYSTEM CALLS .

```
cosmic@COSMIC: /mnt/c/Use x + v
Windows PowerShell
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PS C:\Users\mohan\OneDrive\Desktop\LSP> wsl
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ gedit day1.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ g++ day1.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ ./a.out
Total RAM In MB: 7789 MB
Total RAM In GB: 7 GB
Total unused RAM: 6497MB
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ gedit day1.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ |
```

DAY2: DISPLAY PROCESS LIST WITH CPU AND MEMORY USAGE.

```
cosmic@COSMIC: /mnt/c/Use x + v
Windows PowerShell
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PS C:\Users\mohan\OneDrive\Desktop\LSP> wsl
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ g++ day1.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ ./a.out
Total RAM In MB: 7789 MB
Total RAM In GB: 7 GB
Total unused RAM: 6497MB
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ gedit day2.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ g++ day2.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ ./a.out
User Time :4091
Nice Time :202
System Time :40600
Idle Time :33291655
iowait Time :8143
irq Time :0
softirq Time :7045
steal Time :0
guest Time :0
guest_nice Time :0
idle_time:33299798
Total_CPU_Time:33396736
Active_Time :96938
CPU_Usage0
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ |
```

DAY3: IMPLEMENT PROCESS SORTING BY CPU AND MEMORY USAGE :-

```
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ gedit day3.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ g++ day3.cpp
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ ./a.out
PID      CPU%    Memory (kB)   Name
1         0.0169379    0             (systemd)
54        0.012669    0             (systemd-journal)
97        0.0116456    0             (systemd-udev)
2859     0.00952593   5276          (bash)
163       0.00889301    0             (dbus-daemon)
2515     0.00822346   4460          (dbus-daemon)
1527     0.00579661   25644         (docker-desktop-)
156       0.00529346   6456          (systemd-timesyn)
189       0.00487015    0             (wsl-pro-service)
2852     0.00475541    0             (Relay(2859))
cosmic@COSMIC: /mnt/c/Users/mohan/OneDrive/Desktop/LSP$ |
```

DAY4: ADD FUNCTIONALITY TO KILL PROCESS:-

```
Windows PowerShell
cosmic@COSMIC:/mnt/c/Users/mohan/OneDrive/Desktop/LSP$ g++ day4.cpp
cosmic@COSMIC:/mnt/c/Users/mohan/OneDrive/Desktop/LSP$ ./a.out
PID    CPU%    Memory (kB)    Name
3002   0.108196    5276    (bash)
2995   0.027049     0    (Relay(3002))
1      0.017083     0    (systemd)
54     0.0127001   0    (systemd-journal)
97     0.0116562   0    (systemd-udev)
163    0.00904663  0    (dbus-daemon)
2515   0.00771614  4460  (dbus-daemon)
1527   0.00573717  25644  (docker-desktop-)
156    0.00525401  6456  (systemd-timesyn)
189    0.00494103  0    (wsl-pro-service)
Enter PID to Kill :
3002
Process 3002 terminated successfully
PS C:\Users\mohan\OneDrive\Desktop\LSP> |
```

DAY5: IMPLEMENT REAL-TIME UPDATE FEATURE TO REFRESH DATA EVERY FEW SECONDS:-

```
Windows PowerShell
PID    CPU%    Memory (kB)    Name
3048   0.0216908    5176    (bash)
1      0.0172924     0    (systemd)
54     0.0128013   0    (systemd-journal)
97     0.0116529   0    (systemd-udev)
163    0.00925475  0    (dbus-daemon)
2515   0.00607215  4460  (dbus-daemon)
1527   0.00569559  25644  (docker-desktop-)
3043   0.0054227   0    (Relay(3048))
156    0.00520158  6456  (systemd-timesyn)
189    0.00496531  0    (wsl-pro-service)
Enter PID to Kill :
3048
Process 3048 terminated successfully

Press 'q' to quit or Enter to refresh :
PS C:\Users\mohan\OneDrive\Desktop\LSP> |
```

After refreshing:-

```
cosmic@COSMIC:/mnt/c/Use
PID    CPU%    Memory (kB)    Name
3123   0.004      5232    (bash)
1      0.0172789     0    (systemd)
54     0.0128088   0    (systemd-journal)
97     0.0116993   0    (systemd-udev)
163    0.00924517  0    (dbus-daemon)
2515   0.00583779  4460  (dbus-daemon)
1527   0.00574707  25644  (docker-desktop-)
156    0.00521092  6456  (systemd-timesyn)
189    0.00497575  0    (wsl-pro-service)
458    0.00463986  11384  (systemd)
Enter PID to Kill :
|
```